FLEET TEAM DIGEST

737MAX-FTD-32-17010

Issue Title : MLG HANGER LINK CRACK INDICATIONS IN THERMAL COATING

Airplane Model	ATA	Minor Model(s)
737MAX	3210-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
05/02/2017	05/02/2017	04/21/2017

Estimated Completion Date	Next Update Date

Description

During overhaul of main landing gears, the MLG Hanger Links (or sometimes called Swing Links) are inspected for cracks per CMM 32-11-13 using the penetrant method. During the Non-Destructive Inspection (NDI) procedure, crack indications on the ID were detected on some 737NG aircraft. The crack indications are typically located on two areas (opposite to each other - 180 degrees) of the inner diameter that have F-15.380 coating (.004-.006 thick) applied on the spherical ID. See illustration.

Background

At overhaul, parts are inspected for signs of service-induced cracking. Because Magnetic Particle Inspection (MPI) can not be used for Titanium (Ti) parts, they all must receive a Fluorescent Penetrant Inspection (FPI) which typically also includes the coated surface. Indications in the coating must be reported and further evaluated to ensure that the cracking does not go into the base metal.

The cracking issue appears to be related to the reduced stiffness of Ti substrates relative to steel. Because Ti components flex more under load, the coating is subjected to higher strain levels than on steel parts, which in turn leads to a high incidence of coating cracks. So far, none of the stripped parts displayed indications in the base metal upon reinspection, so no rework was required except to recoat and finish grind.

Status

There currently is no available inspection method to determine if the cracks go through the entire coating thickness and into the titanium substrate. We are working to develop other NDI inspection procedures that could be used to verify that coating cracks have not progressed into the base metal.

Interim Action

IF FPI reveals crack indications in the areas of the coating, then lightly polish using fine diamond impregnated paper and perform the FPI again. If the markings cannot be removed by polishing, then the entire coating should be removed and the base metal reinspected before the thermal coating is reapplied.

Final Action

If cracks are found in these coated Ti parts at overhaul, then it is necessary to strip and recoat per CMM requirements if polishing is not sufficient.

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Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Component Maintenance Manual (CMM)	32-11-13		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
MLG Hanger Link Assy	161A4300-3xx		

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- In Service Occurrences : null

Attachments:

a) Cracks found at overhaul inspection

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